Forecasting Household Appliance Energy Consumption:

A Comparative Analysis of Benchmark, Machine-Learning and Foundation-Model Forecasts

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1. Introduction

This study takes a machine learning approach to answer the essential question of which forecasting method predicts household appliance energy consumption with the least prediction error. The research aims to contribute towards smart energy management by answering forecasting-related queries like demand-side response, battery and storage sizing, and consumer feedback.

The comparison between eight methods is made on the Appliances Energy Prediction of the UCI Machine Learning Repository (Candanedo, Feldheim & Deramaix, 2017), including SARIMAX, tuned XGBoost, the Chronos-2 pre-trained model, and five transparent benchmarks (mean, naive, daily seasonal naive, weekly seasonal naive, and drift). The real-world 24-hour-ahead forecasts over the last two weeks of the dataset were used, including 14 different daily forecasts in total. Besides finding out which method has the lowest prediction error, the study also aims to find out why the performance of different models varies and whether, in terms of realistic forecasting, more complex methods significantly benefit over simpler seasonal benchmarks.

2. About Data and Preprocessing Techniques

The dataset used in this analysis consists of appliance energy use in watt-hours with indoor temperature and humidity from 9 different sensors, lighting and timestamps every ten minutes from January 11 to May 27 of the year 2016, summarised in Table 1 below. As can be seen, the raw file did not contain any missing values, while two columns of artificial variables, rv1 and rv2, were removed for not containing any meaningful information. Additionally, all columns were converted to numeric values, and the data was resampled to an hourly frequency with the use of the mean, with small gaps being filled with time-based interpolation.

Table 1: Dataset and Target Summary

Summary	Value
Total number of observations	19,735 (recorded at 10-minutes interval)
Observations after hourly reshape	3,290
Total number of variables (except random noise: **rv1** & **rv2)**	26
Training period	2,954 hours
Test period	336 hours
Target Standard deviation	81.21 Wh
Target range	28.33 to 608.33 Wh
Target mean	97.78 Wh
Target median	63.33 Wh

The final 14 days of the dataset are held out as the test data, while the remainder are used for model training, with the target variable being '**Appliances**' (Wh).

3. Exploratory Time Series Analysis

Below are visualisations of the hourly appliance energy use across the recorded time period, as well as the mean appliance use in ***Figure 2,*** split by hour and day of the week.

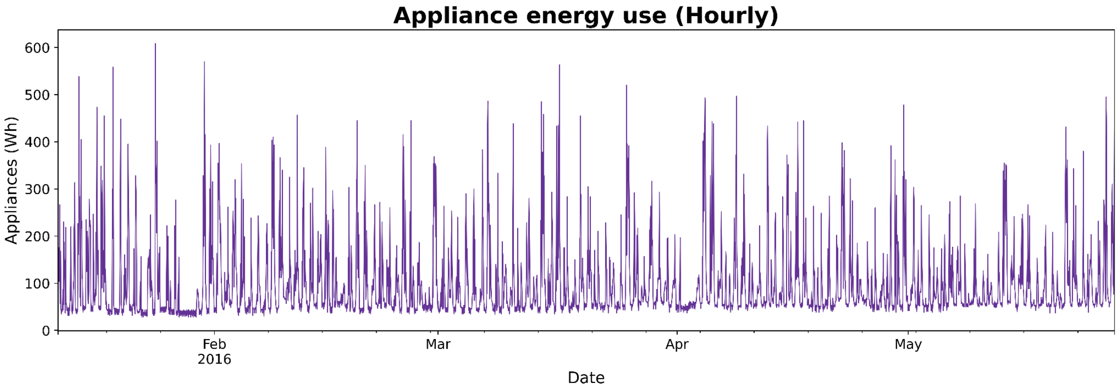

Figure 1: Hourly appliance energy usage.

As can be seen in ***Figure 1***, over the span of 4.5 months, the hourly series does not demonstrate a clear trend but rather exhibits strong within-day seasonality, with large spikes during the day. Notably, within the hour, the power use is highly variable, spiking during the evening (peaking at 192 Wh on average around 6 pm) and dropping to a low (50 Wh on average) overnight to rise sharply again around 6 am. While there is less seasonality within the week, Mondays and Friday/Saturdays tend to exhibit higher power consumption, as can be seen in ***Figure 2*** below.

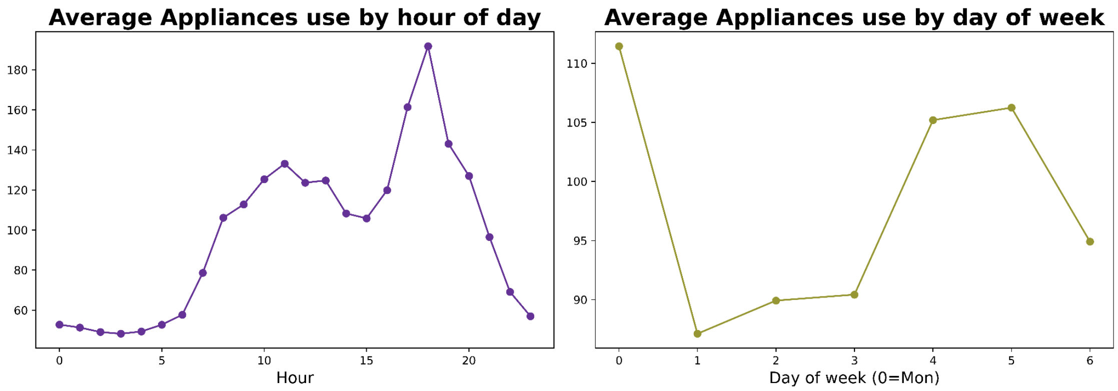

Figure 2: Mean appliance usage (a) by hour, and (b) by day of the week.

Decomposing the time series further, the ACF plot for the series is presented in Figure 3 below, which indicates seasonality at a higher lag order and high serial dependence at a lower lag. Moreover, the series was consistently stationary with no unit root according to the stationarity tests (KPSS and ADF). Specifically, the stationarity null hypothesis was not rejected by the KPSS test (KPSS: 0.038, p = 0.10), while the unit-root null hypothesis was rejected by the ADF test (Dickey & Fuller, 1979) (ADF: -8.95, p < 0.001). Thus, this result suggested that the series was kept in levels, without differentiation.

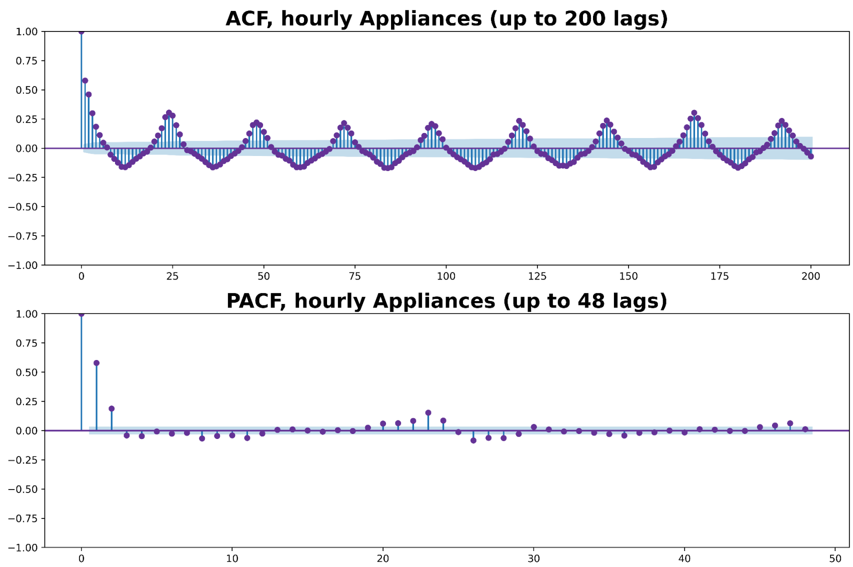

Figure 3: ACF and PACF, hourly Appliances

4. Forecasting Design

The target variable is Appliances (Wh), which is hourly and has a 24-hour forecast horizon. The data used for training amounts to the first 2954 hours, while the last 14 days (336 hours) are held out for testing. Instead of a single multi-step-ahead forecast, the code evaluates 14 separate 24-hour-ahead forecasts, thus simulating the production environment more realistically.

The evaluation metrics include MAE, RMSE, MASE, and Bias. The latter is used to detect systematic forecast bias, while MASE is a relative metric that allows understanding performance advantages of one forecast over another in terms of the reduction of absolute errors compared to a simple in-sample seasonal naive forecast (Hyndman & Koehler, 2006). In this case, the best-performing model is selected with respect to each metric.

Finally, note that the evaluation design is realistic in terms of operational performance due to using a rolling forecast design, which limits the amount of error carry-over between consecutive forecasts and evaluates forecast accuracy in terms of the model's ability to predict the next 24 hours given whatever information is available then.

5. Benchmark Models

Drift (linear interpolation between the first and last training observations), mean (the mean of the training data), naive (last observation), daily seasonal naive (observation of the same hour one day ago), and weekly seasonal naive (observation of the same hour one week ago) are the five benchmarks used. Since the horizon of 24 hours is smaller than both seasonal periods, without new observations being made in the meantime, seasonal naive forecasts only ever forecast observations that have already been seen in the training data under the rolling 24-hour design.

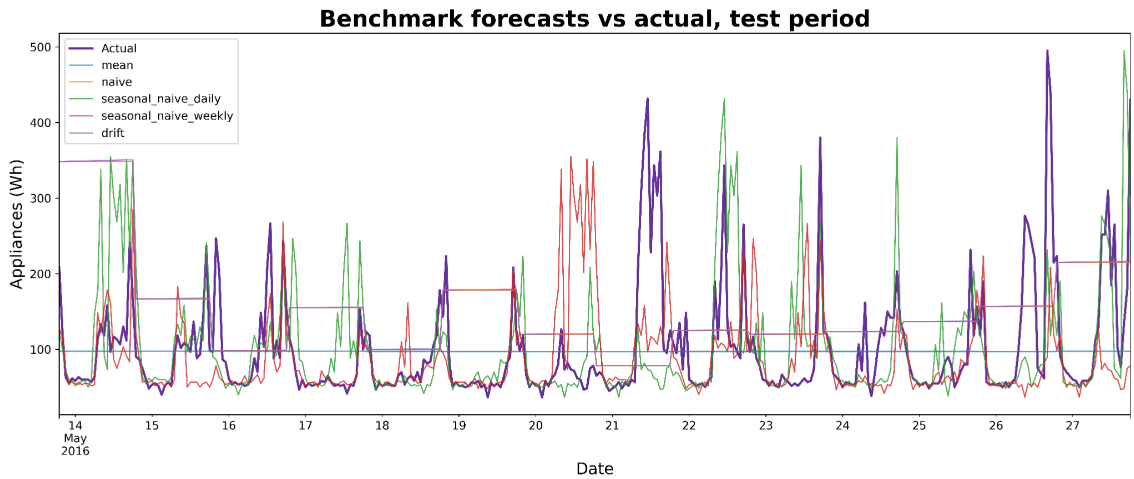

Figure 4: All benchmark forecasts against the observed test period values.

The best benchmark by a reasonable margin was the **weekly seasonal naive forecast** with a **MASE** of **0.813**, just barely edging out the *daily seasonal naive forecast* at *0.904*. Even though a same-hour-yesterday heuristic is already a decent baseline in general, the day-of-week structure (the Monday/weekend bump seen in section 3, EDA, above) encodes a large amount of predictive power not captured by a simple seasonal naive approach, suggesting that appliance use is not strictly following some daily pattern. Linear interpolation between observations (drift) and carrying the mean forward (mean) both fail to account for the daily pattern in the series, performing much worse than seasonal naive.

6. SARIMAX Model

Computational cost was a limiting factor for the order selection using AIC. Since each SARIMAX fit on the 2954-hour training set took between 20 and 100+ seconds, a complete order selection search over 147 combinations was not feasible. Due to stationarity findings in Section 3, the search was limited to p, q $\in$ [0, 3] and d = 0 (16 combinations), while the seasonal order was fixed to (1, 1, 1, 24) as a documented computational limitation. The full order search will be deferred to later.

Five weather variables were included in the SARIMAX(1,0,3)×(1,1,1,24), but none showed significance at the 5% level ($p > 0.17$). Instead, the seasonal MA(1) (-0.958) and AR(1) (0.729) were highly significant, which suggests that the autoregressive and seasonal differencing components were the main drivers of the forecast.

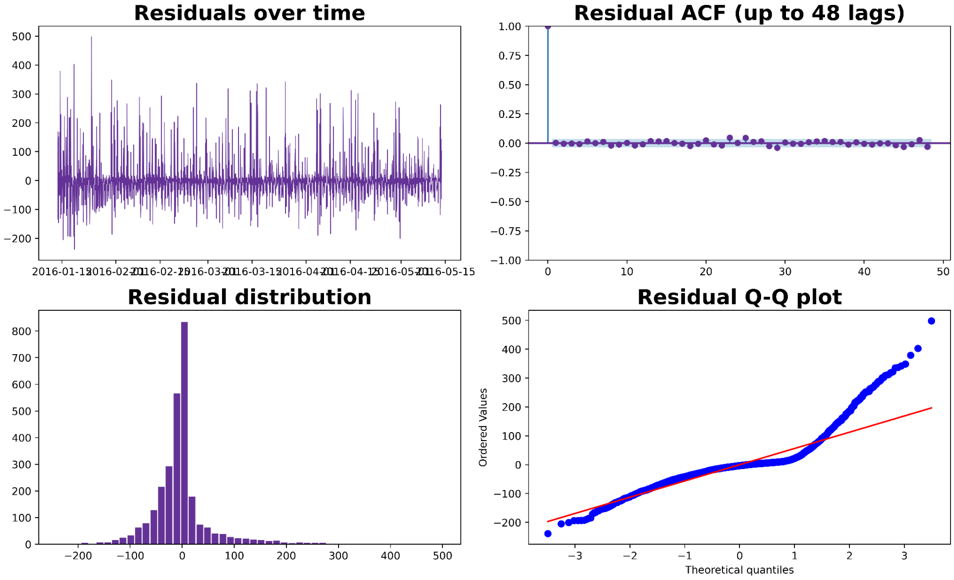

Figure 5: SARIMAX forecast versus actual observation: time series, ACF, distribution, and Q-Q plot.

Since the time series component is supposed to capture the daily cycle, it is unsurprising that the Ljung-Box test failed to reject the null hypothesis of no autocorrelation up to lag 24 ($p = 0.783$ at lag 24, $p = 0.584$ at lag 48); however, the residuals were notably non-normal (skew 1.88, kurtosis 11.18), mostly driven by spikes in the time series that the model was unable to capture with exogenous variables and short-term autocorrelation. In practice, this means that the model's confidence intervals, constructed using the normal distribution for residuals, may underestimate the risk of a high residual occurring, especially around spikes; see Figure 5, where the real observation penetrates the 95% CI band deeper than its $\pm 2\sigma$ width suggests.

When compared to the one-shot 336-hour forecast with the same model, the reconfiguration of the rolling design produced significantly different shapes of the forecasts with only a marginal shift in accuracy (MASE 0.684 → 0.689). The effect size decreases exponentially with increasing horizons ($0.729^2 \approx 0.531$; $0.729^3 \approx 0.387$), and the series exhibits a strong autoregressive component (AR(1) coefficient of 0.729).

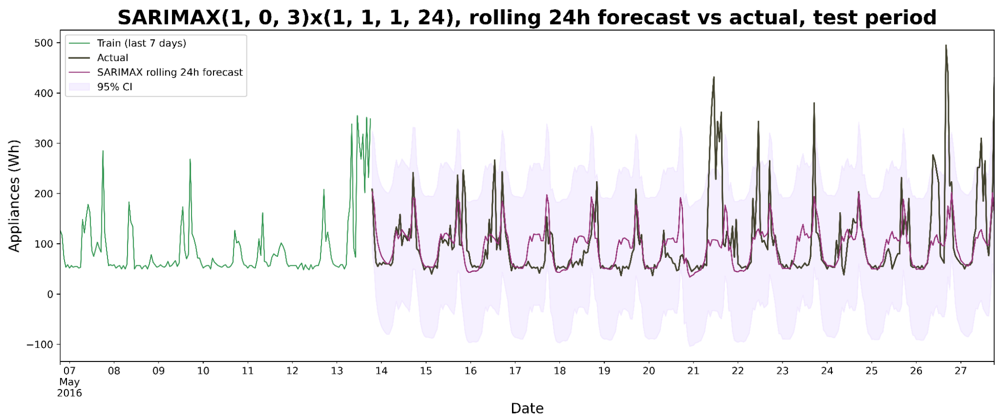

Figure 6: SARIMAX rolling 24-hour forecast.

Thus, I attribute the dampening of day-specific peaks predominantly to the limited expressiveness of the model in capturing day-specific effects through the introduced exogenous variables and short-term autocorrelation rather than the forecast horizon.

7. Feature-Based Model (XGBoost)

Three groups of leakage-free features were engineered for the creation of a 50-feature table: target lags and rolling statistics (lags 1-168 and rolling mean/std), sensor/weather-related variables, and temporal features such as hour, day-of-week, weekend indicators, and their cyclic transformations. Lagged and rolling features were formed only by the previous observations, thus avoiding leakage. Secondly, RandomizedSearchCV with 5-fold TimeSeriesSplit was used for hyperparameter optimisation in order to avoid future leakage that could happen if the usual shuffled k-fold were used. Finally, it was found that regularisation reduced the test MAE by approximately 10%, from 32.46 to 29.26. This was achieved by reducing the number of estimators, max_depth, and learning_rate from 600, 5, and 0.03 to 200, 4, and 0.01. Thus, it can be said that the search for optimal hyperparameters in this case was successful due to the overfitting nature of the manual grid search.

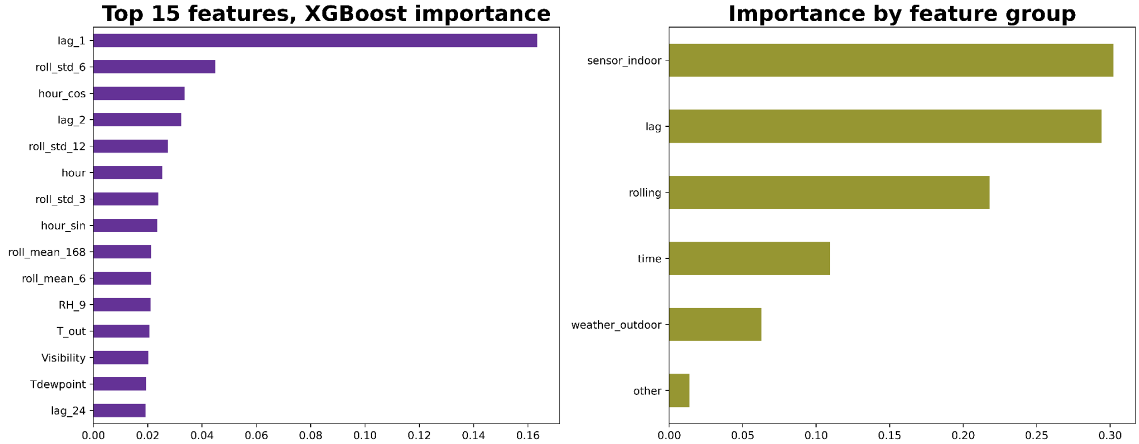

Figure 7: XGBoost feature importance

The energy reading from the prior hour holds the top spot in the individual feature importance rankings, highlighting the strong short-term persistence in the time series. This aligns with findings from the SARIMAX model, where AR(1) emerged as one of the most influential parameters, although the two models capture this dependency differently. Interior sensor data contributes the largest share (0.302), followed by lagged features (0.294), rolling statistics (0.218), time-based variables (0.109), and weather data (0.063). XGBoost's better performance may stem from its ability to integrate recent energy consumption values with supplementary inputs from interior sensors, information not directly available to SARIMAX. With an MASE of 0.548, XGBoost surpassed SARIMAX by 20.5% and improved upon the Weekly Seasonal Naive benchmark by 32.6%.

8. Foundation Model (Chronos-2)

The Chronos-2 target-only zero-shot foundation model (Ansari et al., 2024) did not require any hyperparameter optimisation or feature engineering. For each of the 14 rolling 24-hour forecasts, the mean of the predicted quantiles was taken as the point forecast; each forecast used only the data available at the beginning of the forecasting horizon. With an MASE of 0.614, Chronos-2 proved to be better than SARIMAX (0.689) and all the benchmarks but slightly worse than the tuned XGBoost (0.548), thus ranking second.

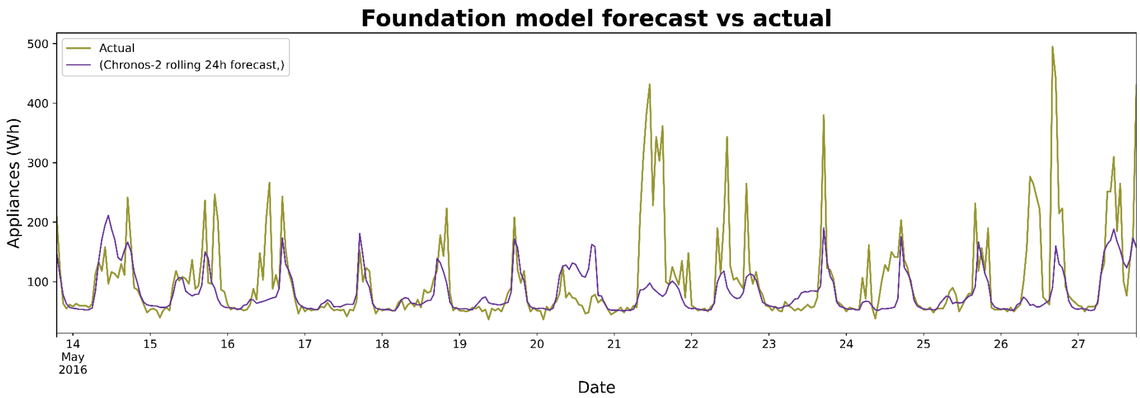

Figure 8: Chronos-2 rolling 24-hour forecast versus actual test-period values.

Chronos-2 had the most negative bias of -18.65 Wh among all the models and the third-lowest RMSE of 67.09. This suggests that it underforecasted the demand more consistently than the other models and had greater extreme values. Chronos-2 smooths out extreme values but generally tracks the day shape of the load quite closely, as can be seen in Figure 8. Chronos-2's overall performance was surprisingly good given the minimal amount of effort required to apply it.

9. Results and Comparison

Table 2 summarises the results of each model over the 336-point rolling 24-hour test period.

Table 2: Evaluation Metrics

Model	MAE (Wh)	RMSE (Wh)	MASE	Bias (Wh)	n
XGBoost	29.26	52.87	0.548	-2.95	336
Chronos-2	32.82	67.09	0.614	-18.65	336
SARIMAX	36.83	65.20	0.689	-5.72	336
Weekly Seasonal Naive	43.36	81.41	0.813	-13.16	336
Daily Seasonal Naive	48.31	85.57	0.904	1.75	336
Mean	50.26	74.94	0.941	-3.29	336
Naive	85.55	110.39	1.601	50.98	336
Drift	85.80	110.68	1.606	51.37	336

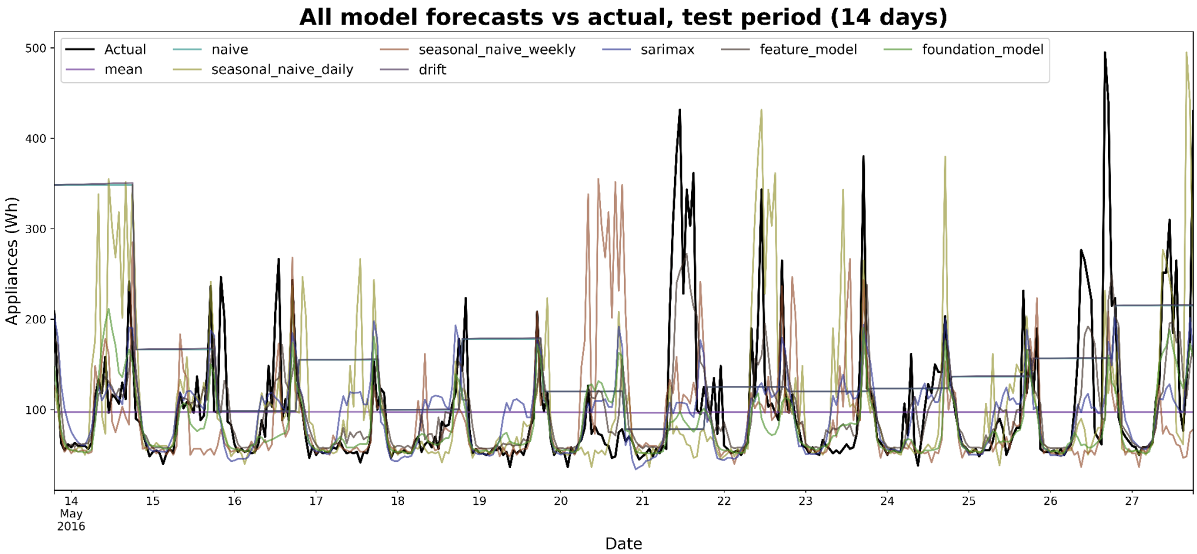

Figure 9: All model forecasts vs actual values.

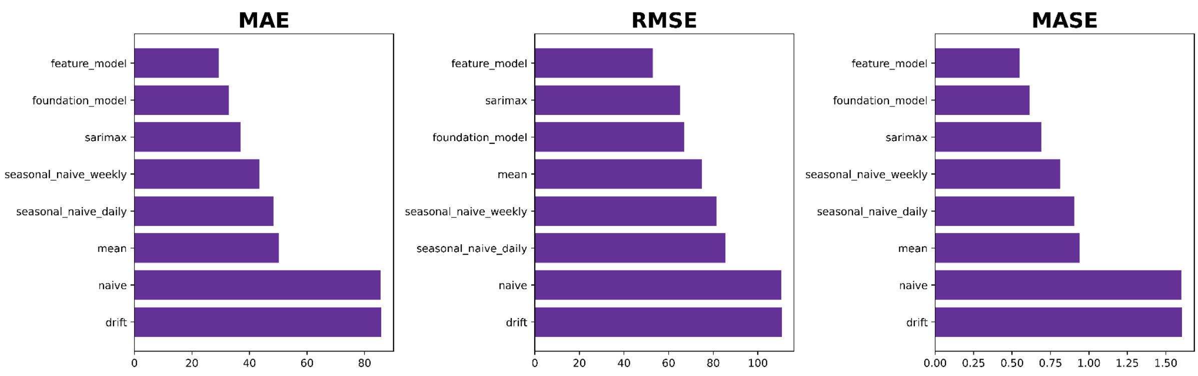

Figure 10: MAE, RMSE, and MASE model ranking.

10. Findings Discussion

Why the results are the way they are

The amount of fresh data available to each model helps explain the ranking in Table 2 above. Weekly seasonal naive is preferred over daily seasonal naive, as expected, because of genuine day-of-week seasonality in appliance use. The value provided by SARIMAX on a 24-hour horizon is diminished because the short-range structure it captures via its ARMA component decays within days. Chronos-2 manages to capture generic day-of-week structure without requiring task-specific engineering, while XGBoost makes better use of sensor variables in its rolling and lagging features to capture near-instantaneous power draw associated with occupancy.

A few results were massively improved by switching to true rolling 24-hour evaluation. Because its previous design involved feeding forecasts into the history, daily seasonal naive benefited most from the switch, with its MASE dropping from an unrealistically high 1.628 to 0.904. By comparison, SARIMAX's MASE only moved slightly (from 0.684 to 0.689) from the change, as its short-range structure was already largely bled out within days. Rolling evaluation did not affect XGBoost's features, as its rolling and lagging design already used proper historical observations. This serves as a good lesson in how evaluation design can unfairly penalise certain classes of models.

Forecasts compared to actual data in the test period

After eliminating the basic heuristics, Figures 4, 6, 8, & 9 confirm that although all the considered models track the overall pattern of the daily electricity profile quite closely, they all significantly underestimate the highest peaks on May 21 and May 26-27. Thus, it was revealed that the information contained in the lagged values of appliance energy use, the calendar variables, and the weather does not fully capture the necessary dynamics for short-term load forecasting since there is no way to take into account such a scenario as, for example, simultaneously using an oven and a kettle. As a consequence of this, Figure 5, presenting the residuals diagnostic of the SARIMAX model with skewness of 1.88 and kurtosis of 11.18, confirms the non-normality of the error distribution with a tendency towards extremes.

However, having a lower value of MAE (29.26 Wh) and MASE (0.548) compared to SARIMAX (36.83 Wh; 0.689) and a lower RMSE (52.87 Wh versus 65.20 Wh), XGBoost demonstrates better performance than the classical time series model. Moreover, the slight deviation of the predicted values from the actual ones with the highest peaks underlines the difficulty of forecasting such events, since the higher RMSE compared to MAE indicates the presence of relatively high errors.

The rationale behind important decisions and their effects

- To improve tractability at the expense of exhaustiveness, a restricted SARIMAX grid search was performed (Section 6). The stationary order identified in Section 3 reduces the dimensionality of the search space; if not fully specified, an untested combination outside the reduced search space might have systematically outperformed the ones that were tried. Thus, the selected order is not fully compliant with brief-requirement forecasts.
- To disentangle the quality of a model from the accuracy of the weather forecasts used as covariates, it was decided to use realised sensor and weather values rather than forecasts thereof for SARIMAX's exogenous inputs and XGBoost's covariates. Chronos-2 has no such option due to its target-only nature, so its 0.614 is not affected by this caveat, which is actually more important than the mere difference in MASE between it and XGBoost indicates (see deployment recommendation below). Thus, both models' accuracy scores are conditional forecasts given access to certain information, not strictly deployable ones.
- In conformity with the leakage avoidance principle used elsewhere in this project, XGBoost used TimeSeriesSplit cross-validation rather than conventional k-fold to avoid fitting the model on data that would otherwise be in the future relative to the validation set. As a result, the model was actually improved by approximately 10% MAE on the held-out test set as compared to what the conventional cross-validation would have selected.

Deployment recommendation

As emphasised by the problem statement, the choice between these models should consider a variety of factors beyond pure forecast accuracy. For instance, constraining XGBoost to only the truly known features (calendar time + the appliance target's own lags/rolling features), as one would have to with Chronos-2, may well eliminate most of its advantage over the transformer-based model (0.548 MASE vs. 0.614), while Chronos-2 achieved its high accuracy with virtually no training or infrastructure investment. Practically speaking, Chronos-2 should be used as the primary model in an honestly deployable forecast, with XGBoost (constrained to the right features) used as a secondary model for comparison and SARIMAX retained as an inexpensive sanity check with proper confidence intervals for interpretability, despite its significantly worse accuracy.

11. Limitations and Future Work

- Extend the search for SARIMAX: Since the reduced search may have failed to find the optimal model order, try the full parameter grid with more computational resources.
- Use forecasted weather: To sidestep the conditional-forecast limitation for the SARIMAX and XGBoost methods, substitute the observed weather with the day-ahead forecast and account for the forecast error.
- Add the uncertainty estimates from XGBoost: Similarly to how the SARIMAX and Chronos-2 intervals were obtained, prediction intervals from the XGBoost model could be found using quantile regression (Koenker & Bassett, 1978) or conformal prediction.
- Examine other foundation models: To verify if Chronos-2's strong zero-shot performance generalises to other foundation models, try the same experiment with alternative models such as TimesFM or TimeGPT. (cf. Makridakis, Spiliotis & Assimakopoulos, 2022)
- Examine different model combinations: Due to the differing error characteristics of the two methods, consider combining the forecasts of XGBoost and Chronos-2 using model averaging or stacking to achieve higher accuracy than each method individually.

12. Conclusion

In this case, complexity paid off, albeit unevenly: most of the exploitable structure in the series was captured by the simplest possible weekly seasonal naive (MASE 0.813), with SARIMAX adding another 15% by modelling additional short-range dependence in the series, though that proved to be a limited advantage due to the decay of short-range memory within a day. XGBoost added more value, largely by exploiting its edge of indoor sensors, albeit with the caveat of being conditional on the knowledge of future events; Chronos-2 matched that advantage with no special tuning required whatsoever and represents a nearly effortless forecast with genuinely deployable accuracy. Combined with an inexpensive but informative SARIMAX sanity check with confidence intervals, which itself built upon the same cheap seasonal naive, that makes for a more compelling picture than raw accuracy rankings suggest and is potentially a viable solution for a realistic smart-home scenario.

13. Reference

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